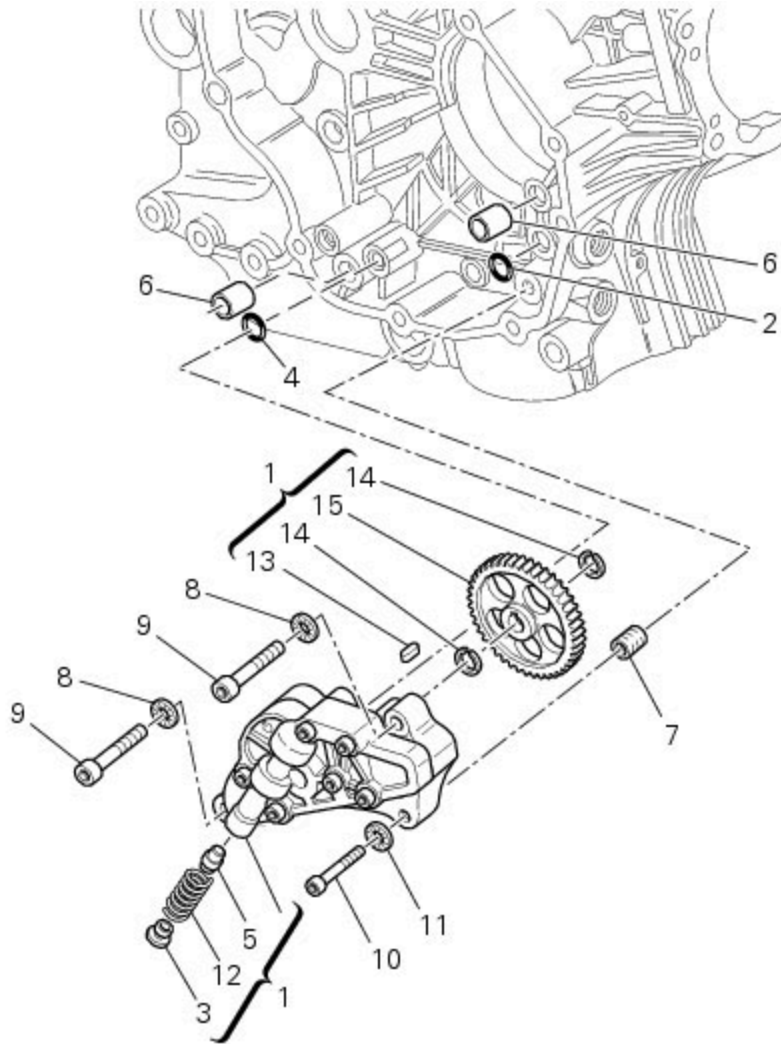


2.1 -Lubrication system: oil pump



- 1 Oil pump complete with gear
- 2 O-ring
- 3 By-pass plug
- 4 O-ring
- 5 By-pass valve
- 6 Locating bush
- 7 Reducer bush
- 8 Spring washer
- 9 Screw
- 10 Screw
- 11 Spring washer
- 12 By-pass spring
- 13 Key
- 14 Circlip
- 15 Pump drive gear

 Spare parts catalogue

1100 ABS [FILTERS AND OIL PUMP](#)

1100 S ABS [FILTERS AND OIL PUMP](#)

Important

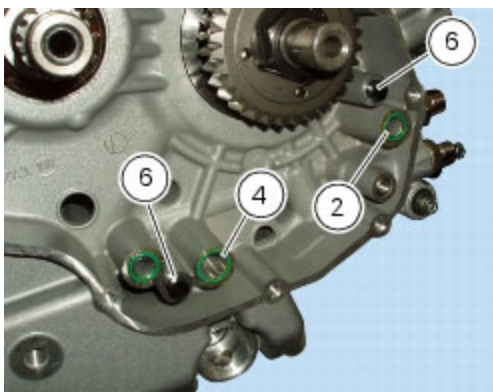
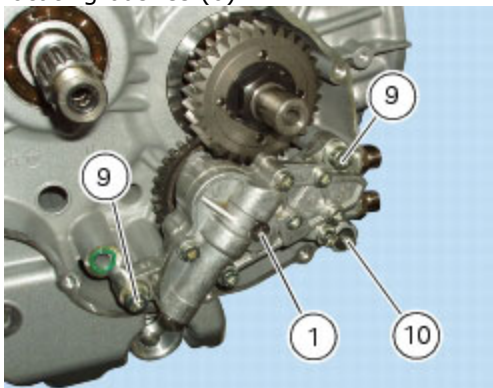
Bold reference numbers in this section identify parts not shown in the figures alongside the text, but which can be found in the exploded view diagram.

Removal of the oil pump

Operation	Section reference
Drain the engine oil	D 4, Changing the engine oil and filter cartridge
Remove the clutch centre	N 6.1, Removal of the clutch
Disconnect the oil pressure sensor	P 5, Checking the indicating devices
Remove the clutch cover	N 6.2, Removal of the clutch cover

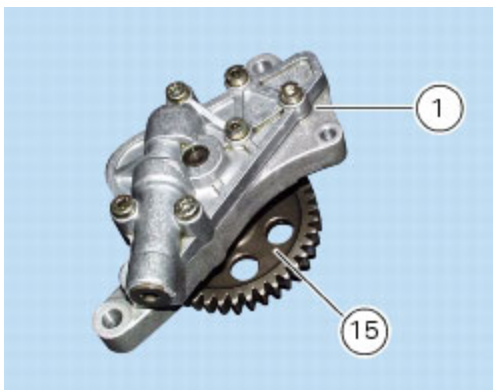
Undo and remove the screws (9) and (10) securing the pump assembly.

Remove the oil pump assembly (1) and extract the O-rings (2) and (4) from the crankcase half together with two locating bushes (6).



Disassembly of the oil pump

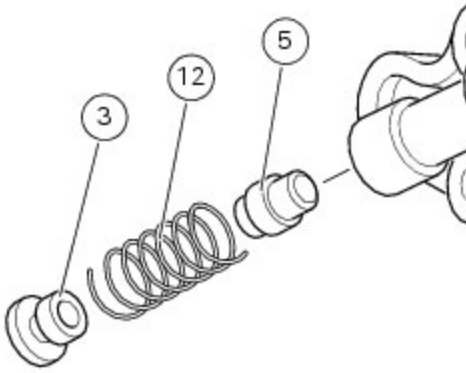
Hold the oil pump (1) in a vice taking care not to damage the drive gear (15).



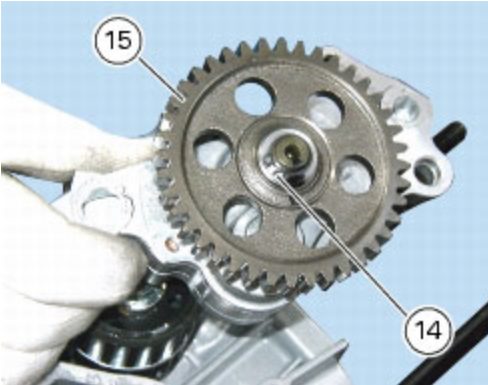
Warning

Make sure that vice jaws are faced with soft material.

Remove the plug (3) and extract the spring (12) and by-pass valve (5).
Check the condition of the above components.

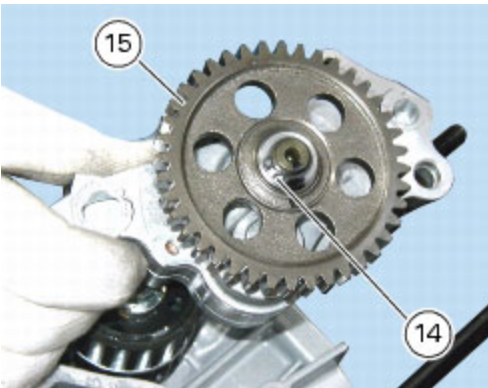


Check the condition of the above components.
 Remove the circlip (14).
 Slide out the pump drive gear (15).

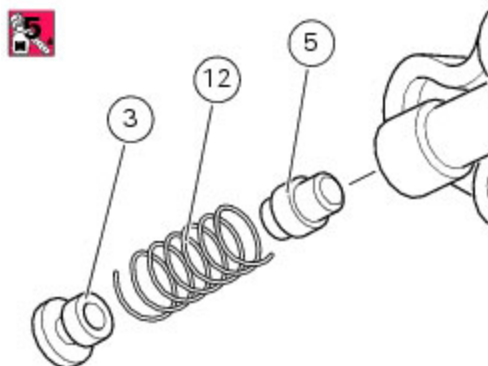


Reassembly of the oil pump

Fit the pump drive gear (15) on the oil pump and secure it by installing the circlip (14) in its seat.

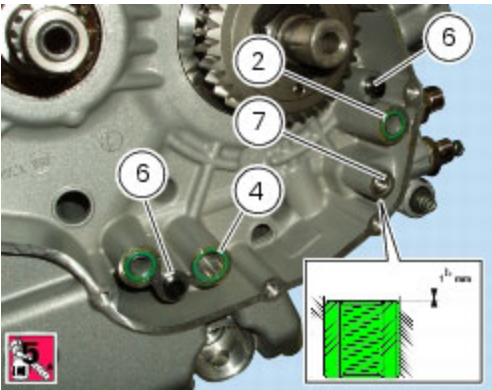


Insert in the pump the by-pass valve (5), the spring (12) and screw and tighten the plug (3) to the specified torque (Sect. C 3, [Engine torque settings](#)) by applying a threadlocker of medium strength.

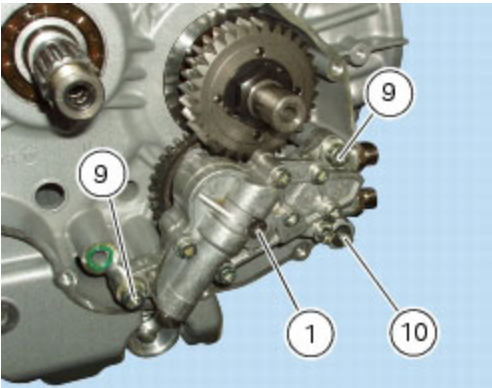


Refitting the oil pump

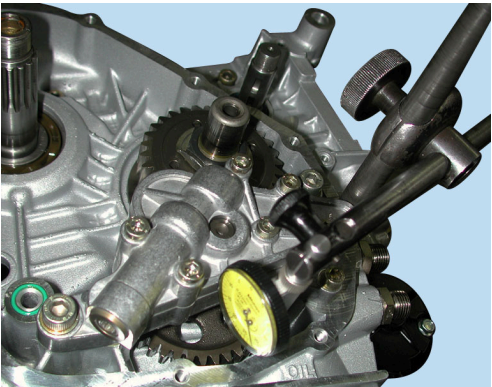
If removed, apply specific threadlocker on the bushing (7) and screw it in the crankcase half, observing the height. Position the reference bushings (6) and the oil sealing O-rings (2) and (4) according to the crankcase lubrication channels.



Position the oil pump (1) on the crankcase and tighten the screws (9) and (10) to the specified torque (Sect. C 3, [Engine torque settings](#)).



Check the gear backlash with the driving pinion by fixing a dial gauge, equipped with the relevant stylus, to the crankcase half.



Position the dial gauge stylus on one tooth of oil pump gear and set the gauge to zero in this position. Move the gear slightly to measure the backlash; take four readings in diametrically opposed positions of the gear. The clearance should be **0.10** mm.

Operation	Section reference
Refit the clutch cover	N 6.2, Refitting the clutch cover
Connect the oil pressure sensor	P 5, Checking the indicating devices
Refit the clutch drum, the clutch centre and the clutch plate pack	N 6.1, Reassembling the clutch
Top up the engine oil	D 4, Changing the engine oil and filter cartridge